MANTHAN SHRIRAM JOSHI

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Education

Northeastern University, Boston | MA | *Master of Science in Cell and Gene Therapies* *December 2026*
Coursework: Stem Cells and Regeneration, Cell and Gene Therapies, Immunology, Regulatory Landscapes, Advanced Drug Delivery Systems, Bioinformatics, Scientific Communication and Biotech Enterprise

Manipal Institute of Technology (MAHE), Manipal | India | *Bachelor of Technology in Biotechnology* | Minor: Pharmaceutical Biotechnology *July 2023*
Coursework: Genetics, Immunology, Bioinformatics, Pharmaceutical Science, Microbiology, Biochemistry, Bioprocess Engineering

Harvard Business School, Boston | MA | *Certificate in Management Essentials* *August 2025*
8-week, 35-hour program emphasizing decision-making, implementation, organizational learning, and change management

Skills

Cell Culture & Aseptic Technique	Aseptic handling and maintenance of mammalian cell lines (CHO) across 5+ passages at >80% viability; cryopreservation and recovery; drug-selection optimization (puromycin, 6-thioguanine); kill curve design using CellTiter-Glo 2.0 cytotoxicity assays across serial dilutions; mammalian cell preparation for 3D bioink formulation; single-cell suspension preparation for flow cytometry.
Genome Engineering	End-to-end CRISPR/Cas9 knock-in and knock-out workflow design; Cas9-RNP complex assembly with HPRT1-targeting sgRNAs; CRISPRmax lipofection in 24-well format; transient transfection; lentiviral vector handling and titer determination.
Plasmid & Cloning	Bacterial transformation, colony screening, miniprep purification; restriction-enzyme digestion (NotI, EcoRV) for construct verification; validation of multiple plasmid constructs by gel readout.
PCR, Sequencing & Validation	Genomic DNA extraction; Phusion PCR amplification of target loci; primer design in Benchling for diagnostic amplicons (600–800 bp) and knock-in confirmation primers flanking homology arms; Sanger sequencing submission and ICE indel analysis; agarose gel electrophoresis; Nanodrop quantification and purity assessment.
Protein Biochemistry	Recombinant membrane protein expression and purification by Ni-NTA affinity chromatography (IMAC); SDS-PAGE; Western blotting; Coomassie staining; ELISA; cytotoxicity assays.
Imaging & Software	Fluorescence microscopy for transgene confirmation at defined intervals; Benchling (primer design, construct mapping); GraphPad Prism (dose-response curve fitting); ICE web tool; R (basic).

Professional Experience

Northeastern University, *Teaching Assistant - Cell and Gene Therapies*, Boston, MA *Jan 2026 - Apr 2026*
- Reinforced graduate-level CGT concepts in office hours and discussion sessions, integrating recent literature on CAR-T therapy and iPSC-to-T-cell differentiation.
- Reviewed and provided structured written feedback on student project deliverables; conducted in-depth literature review of allogeneic and autologous CAR-T platforms to support classroom case discussions.

Northeastern University, *Graduate Research Project*, Boston, MA *Sept 2025 - Dec 2025*
- Maintained CHO cell cultures across 5+ passages at >80% viability; assembled and delivered Cas9-RNP complexes with 2 HPRT1-targeting sgRNAs via CRISPRmax lipofection in 24-well plates for an end-to-end HPRT1 knockout/knock-in simulation.
- Ran CellTiter-Glo 2.0 cytotoxicity assays across 8 serial 3-fold dilutions of 6-thioguanine, graphed dose-response in GraphPad Prism, and selected 100 μM ($\sim$7% viability) to enrich HPRT1-KO cells; confirmed by fluorescence microscopy and prepared single-cell suspensions for flow cytometry.
- Extracted gDNA from 6-TG-selected pellets, amplified HPRT1 by Phusion PCR, and submitted 8 amplicons for Sanger sequencing and ICE indel analysis to confirm gene disruption.
- Validated 5 plasmid constructs by NotI/EcoRV restriction digest and gel electrophoresis; designed 2 primer sets in Benchling (600–800 bp diagnostic amplicon and knock-in confirmation primers flanking homology arms).

Reagene Biosciences Pvt Ltd, *Research Intern*, Bangalore, India *Aug 2023 – Aug 2025*
- Performed core molecular biology workflows including plasmid DNA purification, bacterial transformation, and agarose gel electrophoresis, with Nanodrop quantification for nucleic-acid yield and purity.
- Characterized proteins by Western blotting and ELISA to confirm expression and quantify target proteins.
- Cultured and cryopreserved mammalian cell lines at high viability to supply cells for bioink preparation and 3D-bioprinted human tissue models developed as alternatives to animal testing.
- Contributed to bioink formulation, market research, investor decks, and grant proposals supporting the non-animal testing platform.

Tranalab Pvt Ltd, Center for Human Genetics, *Undergraduate Research Intern*, Bangalore, India *Jan 2023 – Aug 2023*
- Expressed M4, a 35 kDa recombinant membrane protein domain, in *E. coli* BL21(DE3) under IPTG induction as part of a first-ranked undergraduate thesis (top of the 2019 cohort).
- Engineered a modified solubilization buffer by adding 1% Triton X-100 and 1% Tween 20 to an 8 M urea base, increasing soluble protein yield $\sim$48% (A280 0.46 to 0.68) over the standard condition.
- Purified His-tagged M4 by Ni-NTA affinity chromatography (IMAC) using a stepwise urea wash gradient (6, 4, 2 M) and 500 mM imidazole elution; lysed cells by probe sonication and confirmed identity by SDS-PAGE with Coomassie staining.
- Quantified protein by Nanodrop (A280), buffer-exchanged by overnight dialysis, and evaluated refolding by sugar-affinity chromatography.

Indian Institute of Technology, *Undergraduate Research Intern*, Indore, India *Sept 2021 – Oct 2021*
- Supported a project on stem cell-based tissue growth and tumor control through protocol design, MSDS preparation, and experimental planning.
- Assisted with target identification and therapeutic intervention modeling.

Leadership and Volunteering

Vice President, Northeastern University, *Graduate Biotech-Bioinfo Association*, Boston, MA *December 2024 - January 2026*
- Led executive board operations, managing communications, records, and coordination across committees while supporting professional development events and member engagement.
- Oversaw organizational strategy, planning and executing networking events, workshops, and panel discussions to strengthen community involvement and industry exposure.